

Product Recommendation System

Problem Statement & Challenges

Problem Statement

This project aims to build a machine learning-based recommendation system that helps users discover relevant products from a large catalog using historical user ratings. The system implements Popularity-Based Recommendation, User-Based Collaborative Filtering, and Matrix Factorization (SVD) to generate personalized recommendations.

Key Challenges

Cold Start Problem: New users and newly added products have little or no interaction history, making personalized recommendations difficult.

Data Sparsity: Most users rate only a few products, creating a sparse user-item matrix that affects recommendation quality.

Scalability: Computing user similarities becomes expensive as the dataset grows.

Popularity Bias: Highly rated products may dominate recommendations while niche products remain under-recommended.

Solution

To address these challenges, this project combines a Popularity-Based model for general recommendations, User-Based Collaborative Filtering for personalization, and SVD to learn latent relationships and improve recommendations on sparse data.

Future Scope

Hybrid recommendation models, Content-Based Filtering, Precision@K, Recall@K, NDCG evaluation, and deployment using Streamlit or FastAPI.